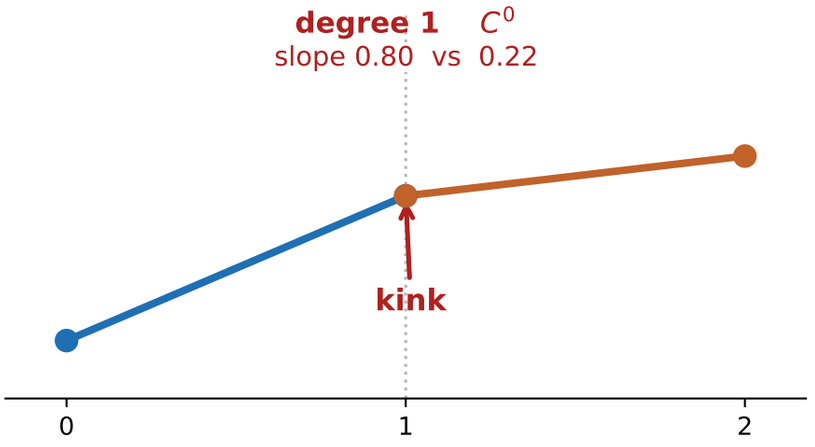
degree and continuity are two different knobs

(a) to (c) turns the DEGREE knob --- the curve gets wigglier inside each element, and the kink never leaves.

(c) and (d) are BOTH degree 3. Only (d) makes the slope a nodal unknown, so the two elements read the same one.

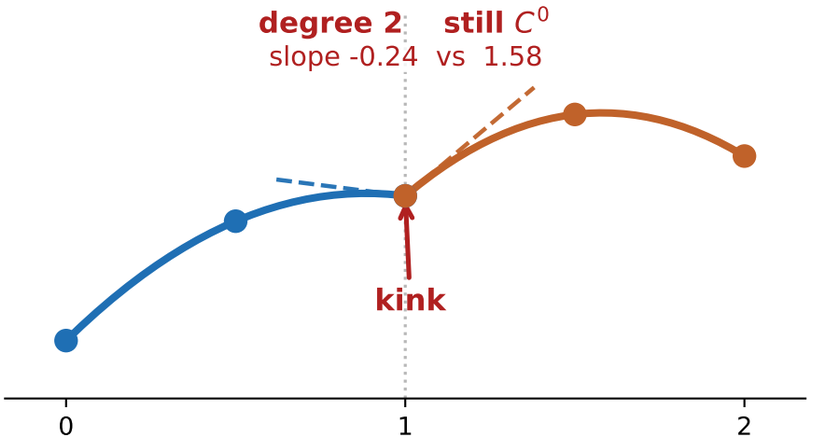
(a) LINEAR Lagrange



$N = \frac{1-\xi}{2},\ \frac{1+\xi}{2}$ (2 nodes)

shared at the node: the value

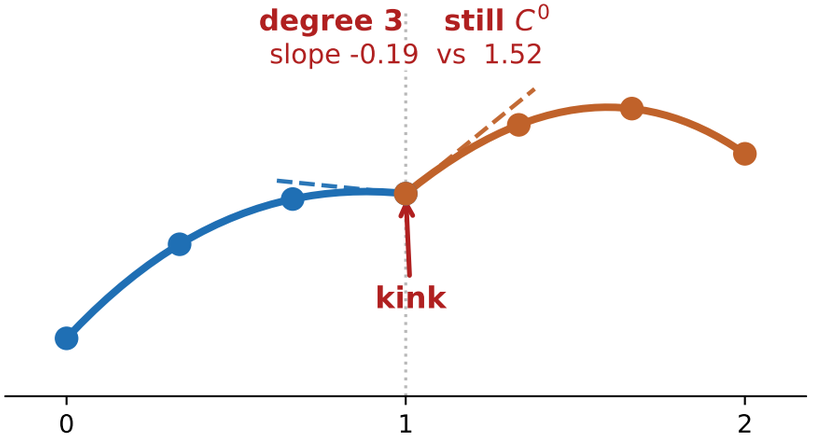
(b) QUADRATIC Lagrange



$N = \frac{\xi(\xi-1)}{2},\ 1-\xi^2,\ \frac{\xi(\xi+1)}{2}$ (3 nodes)

shared at the node: the value

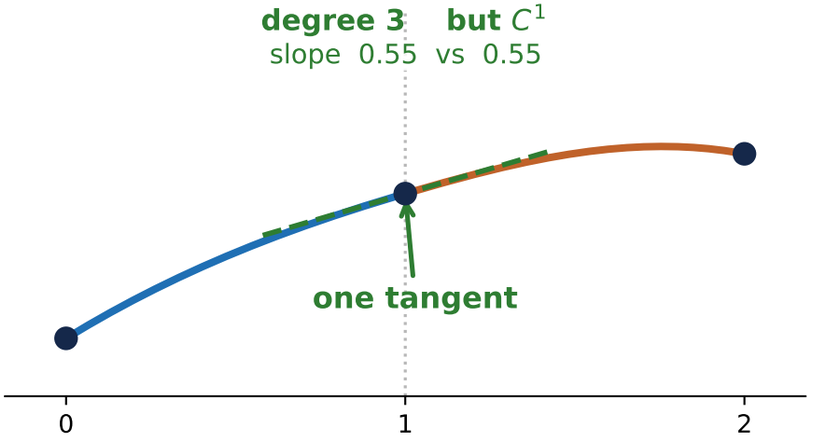
(c) CUBIC Lagrange



4 nodes at $\xi = -1, -\frac{1}{3}, \frac{1}{3}, 1$

shared at the node: the value

(d) CUBIC Hermite



$H = 1 - 3s^2 + 2s^3,\ L(s - 2s^2 + s^3),\ \ldots$ (2 nodes)

shared at the node: the value AND the slope

	(a) linear C^0	(b) quadratic C^0	(c) cubic C^0	(d) Hermite C^1
Poisson, order 2 $\int k u'v'$ needs $u \in H^1$ -> C^0	✓	✓	✓	✓
Euler-Bernoulli beam, order 4 $\int EI\, w''v''$ needs $w \in H^2$ -> C^1	✗	✗	✗	✓

turning the DEGREE knob, (a) to (c), changes nothing in either row.
Only the CONTINUITY knob, (d), does.